



PHANTOM 4.8 // DIAGRAM 12A — MASTER COMPONENT & API GATEWAY INTEGRATION MAP

GUJARAT POLICE DEPARTMENT • NETRAM COMMAND & CONTROL

ACTIVE C2
EXHIBIT

FASTAPI V1 ASYNC API GATEWAY // 23 LOGICALLY GROUPED ROUTERS

Middleware Pipeline: CORS Policy • SSRF Metadata Filter • SlowAPI Rate Limiting • Correlation ID • Audit Logger
Async SQLAlchemy Session Pooling • Pydantic v2 Schema Validation • OpenAPI 3.1 Interactive Docs

AUTHENTICATION & USER RBAC /api/v1/auth & /api/v1/users

JWT tokens, Argon2id, 7-role granular authorization matrix, session revoke

CAMERA DISCOVERY & REGISTRY /api/v1/cameras

Sentinel /api/ingest sync, Corp8SourceAdapter, health ping, CRUD camera records

VIDEO STREAMING GATEWAY /api/v1/streams

WebRTC WHEP preview, RTSP over TCP proxy, HLS fallback, load pacing profiles

ALERT ENGINE & WATCHLISTS /api/v1/alerts & /api/v1/watchlists

60s anti-storm cooldown, severity scoring, plate registration, vehicle alerts

INCIDENT FSM & AUDIT LOGS /api/v1/incidents & /api/v1/audit

FSM: OPEN -> CLOSED, officer notes, tamper-evident Section 65B audit trail

FORENSICS & SIGHTING GRAPH /api/v1/investigations & /api/v1/dossiers

Plate & make search, SHA-256 evidence verification, certified PDF dossier

GIS MAP & TRAJECTORY /api/v1/gis

CCTV FOV coverage wedges, PostGIS geometries, multi-camera intercept prediction

AI COPILOT AGENT INTERFACE /api/v1/copilot

Multilingual conversational agent (EN, HI, GU) with 40+ deterministic grounded tools

TECHNICAL SPECIFICATIONS & ARCHITECTURE MECHANISMS:

- Maps 23 FastAPI routers into 7 functional clusters: Auth, Cameras, Streams, Alerts, Incidents, Investigations, GIS, Copilot.
- Non-blocking async request lifecycle with Pydantic v2 validation and dependency injection for authorization.
- 100% test pass rate across all unit, security, integration, and AI vision test suites (168 tests).